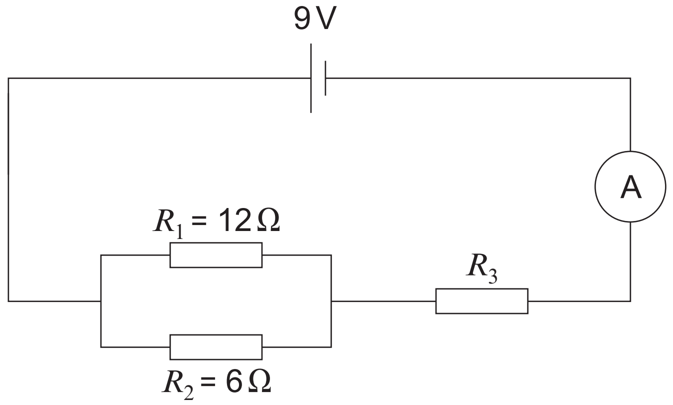
Examiner only

7. A group of students set up the following circuit.



(a) (i) Use the equation:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

to calculate the total resistance of resistors R_1 and R_2 in parallel. [3]

total resistance of R_1 and R_2 = ... Ω

(ii) The total resistance of the circuit is 6Ω.

Use an equation from page 2 to calculate the resistance of resistor R_3. [1]

resistance of R_3 = ... Ω

(iii) Use an equation from page 2 to calculate the current through the ammeter. [2]

current = ... A

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(iv) Calculate the voltage across the resistor R_3. [3]

voltage across R_3 = ... V

(b) One of the students, Katrina, connects a different circuit, without R_1 and R_2.

The circuit only contains R_3 in series with the battery.

She correctly calculates that the current from the battery would be 4.5 A.

She **claims** that the new circuit would transfer 45 J of energy in 10 s.

By using equations from page 2, explain whether her claim is correct. [3]

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